

AR 511: Autonomous Mobile Robots

Course Code	AR 511
Course Name	Autonomous Mobile Robots
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Robot locomotion:** Types of locomotion using hopping, leg, wheel. Types of mobile robots: ground robots (wheeled and legged robots), aerial robots, underwater robots, and water surface robots. Discussion on stability, maneuverability, and controllability. (4 hours)
- **Robot kinematics and dynamics:** Forward and inverse kinematics, degree of freedom and maneuverability, holonomic and nonholonomic constraints, kinematic models of wheeled and legged robots, dynamics simulation, Classification of models, Rigid body dynamics, Lagrange-Euler and Newton-Euler methods. Computer-based dynamic (numerical) simulation of different robots. (12 hours)
- **Sensors for mobile robot navigation:** Proprioceptive/Exteroceptive and passive/active sensors, performance measures of sensors, sensors for mobile robots like global positioning system (GPS), position sensor, gyroscope, accelerometer, magnetic compass, inclinometer, Doppler effect-based sensors, laser scanner, infrared rangefinder, visual and motion; uncertainty in sensing and filtering. (6 hours)
- **Navigation:** Localization, error propagation model, Probabilistic map-based localization, Autonomous map building, Simultaneous localization and mapping (SLAM). (6 hours)
- **Motion and path planning:** Line of sight guidance strategies, Collision free path planning, sensor-based obstacle avoidance, and trajectory tracking. Path planning algorithms based on A-star, Dijkstra, Voronoi diagrams, and probabilistic roadmaps (PRM), rapidly exploring random trees (RRT), Markov Decision Processes (MDP), and stochastic dynamic programming (SDP). (8 hours)
- **Modern mobile robots:** Swarm systems, Cooperative and collaborative systems, and autonomous mobile manipulation. (4 hours)
- **Final project:** Student project towards design, fabrication, and programming of a mobile robot. (4 hours)

Text Books

1. Siegwart, Roland, Illah Reza Nourbakhsh, and Davide Scaramuzza, **Introduction to autonomous mobile robots**, MIT press, 2011.
2. Dudek, Gregory, and Michael Jenkin, **Computational principles of mobile robotics**, Cambridge university press, 2010.
3. Melgar, E. R., Diez, C. C., **Arduino and Kinect Projects: Design, Build, Blow Their Minds**.

References

1. Peter Corke, Robotics, **Vision and Control: Fundamental Algorithms in MATLAB**, Springer Tracts in Advanced Robotics.
2. S. M. LaValle, **Planning Algorithms**, Cambridge University Press, 2006. (Online)
3. Thrun, S., Burgard, W., and Fox, D., **Probabilistic Robotics**, MIT Press.
4. H. Choset, K. M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, L. E. Kavraki, and S. Thrun, **Principles of Robot Motion: Theory, Algorithms and Implementations**, PHI Ltd.